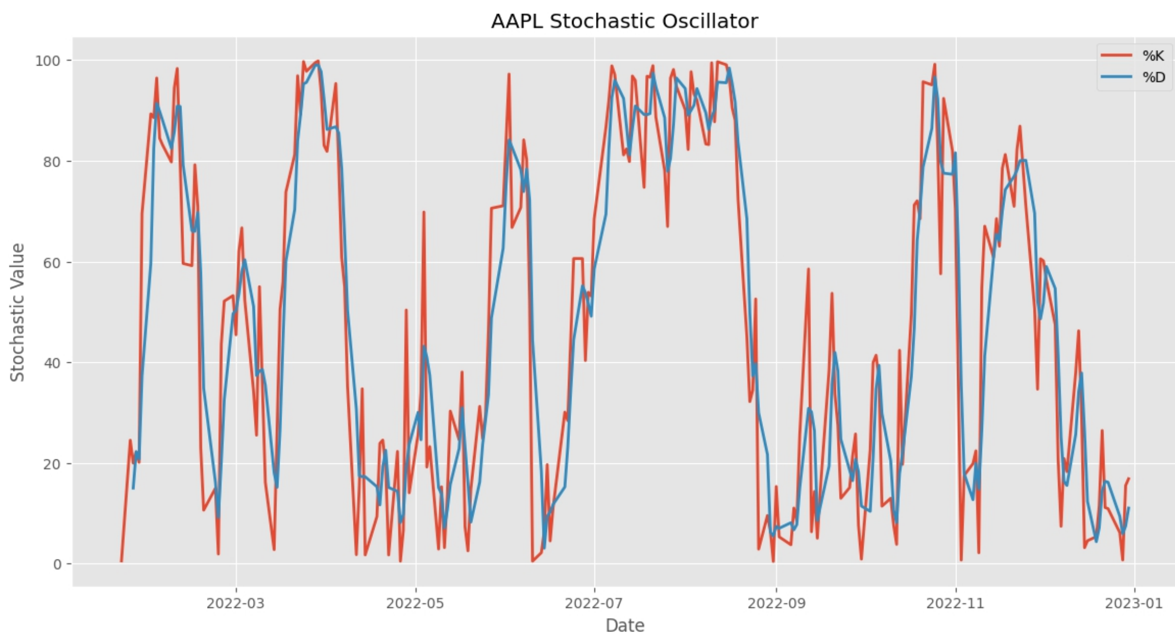
```

    # Calculate %D (slow stochastic) - 3-day SMA of %K
    d = k.rolling(window=d_period).mean()

    return k, d

# Calculate Stochastic Oscillator
apple['Stochastic_K'], apple['Stochastic_D'] = calculate_stochastic(apple)

```



The Stochastic Oscillator generates signals through:

- **Values above 80**: Potentially overbought conditions
- **Values below 20**: Potentially oversold conditions
- **%K crossing above %D**: Bullish signal
- **%K crossing below %D**: Bearish signal
- **Divergence with price**: Potential reversal signal